

SOHAM DAS

Backend Engineer · Python / FastAPI / Distributed Systems

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SUMMARY

Backend engineer with hands-on production experience shipping systems solo: authentication, caching, event pipelines, and LLM-integrated automation. Built and deployed three live systems, a P2P payment wallet with idempotent transfers, a Kafka-based telemetry pipeline sustaining 200 events/sec, and a full agency operations platform with automated AI-driven lead outreach. Seeking a full-time backend engineering role where system design and correctness are taken seriously.

TECHNICAL SKILLS

- **Backend & Frameworks:** Python · FastAPI · Flask · Uvicorn
- **Databases:** PostgreSQL (NeonDB) · MongoDB · Redis
- **Infra & DevOps:** Docker · AWS · GCP · Git/GitHub · CI/CD (GitHub Actions)
- **Backend Concepts:** JWT Auth · Idempotency · Caching · Circuit Breakers · REST API Design · Async Programming · Microservices
- **AI Integration & Automation:** LLM pipeline design (Claude, Gemini) · n8n workflow automation

EXPERIENCE

Backend Engineer — Internship · GameGenesis

Jan 2025 – Mar 2025

Owned all technical decisions, architecture, and production systems end-to-end at an early-stage startup.

- Engineered and scaled core backend services and CRM architecture from scratch, translating business requirements directly into production systems using FastAPI, PostgreSQL/Neon, Redis, and CI/CD via GitHub Actions.
- Integrated LLM pipelines (Claude, Gemini) directly into the CRM, automating workflows that previously required manual handling and cutting manual processing time significantly.
- Built an employee management module to track contribution value per team member, used directly for resourcing decisions.
- Maintained database architecture and all production systems with zero downtime across the full internship period.

PROJECTS

Apex Ledger — Secure P2P Payment Wallet

[GitHub](#) · [Live](#)

Python · FastAPI · PostgreSQL (NeonDB) · Redis · JWT · Docker · pytest · GitHub Actions

- Built a P2P payment wallet supporting user registration, login, and real-time balance transfers with automatic fee calculation and atomic balance updates.
- Implemented stateless JWT authentication via HTTP-only cookies; sender identity is always resolved server-side from the token, never from user input, preventing spoofing.
- Added Redis idempotency keys (300s TTL) to prevent duplicate transfers from double-clicks, plus balance caching (60s TTL) with circuit breaker fallback.
- Secured passwords with bcrypt, enforced SSL on DB connections, containerized with Docker, and configured a full pytest suite running on every push via GitHub Actions.

Gen Argus — Real-Time Telemetry Pipeline

[GitHub](#) · [Live](#)

Flask · Apache Kafka · PostgreSQL (NeonDB) · Pydantic · SQLAlchemy · Docker · Aiven · Render

- Built a telemetry pipeline ingesting events from any client (mobile, web, Postman) into Kafka, batch-flushing to PostgreSQL every 1,000 records or 2 seconds.
- Placed Kafka between API and database to absorb ingestion spikes; sustained 50–200 events/sec on free-tier infrastructure.
- Enforced API key auth on ingestion endpoints and Pydantic schema validation with 422 responses for invalid payloads; live dashboard polls every 3s via Chart.js.
- Deployed as a web service + background worker on Render with Aiven-managed Kafka in production.

Lead Research & Personalization Engine

n8n · Google Sheets API · Google Gemini · Anthropic Claude API · HTTP/REST

- Built an end-to-end automation pipeline in n8n that ingests leads from Google Sheets, filters out already-processed entries, and fetches each company's website content directly via HTTP.
- Feeds scraped content to Gemini (with Claude wired in via direct API calls) to generate structured research output: company summary, core problem, opportunity, personalized email copy, subject line, and a confidence score.
- Writes parsed results back into a separate sheet tab with match-and-update logic keyed on lead name, avoiding duplicate processing.
- Processes leads sequentially with built-in rate-limit delays between API calls, plus isolated error logging so one failed lead never breaks the run.

EDUCATION

B.Sc. Information Technology · Techno India University, Kolkata

Graduating 2026